

Week 3 Task: Monitoring, Logging, and Cloud Security Setup Simulation

Terraform-based AWS monitoring, logging and security implementation

Evidence note: The assignment requires actual screenshots/terminal outputs. This report provides the implementation and evidence checklist but does not fabricate successful deployment evidence.

1. Objective

Build practical monitoring, logging and cloud-security skills by implementing a layered framework for a sample application. Terraform is used to automate the infrastructure and configuration.

2. Architecture

The lab deploys an EC2 web instance and configures CloudWatch Agent for Nginx/system logs plus memory/disk metrics. Security and audit layers include Security Groups, IAM, Systems Manager, VPC Flow Logs, CloudTrail, GuardDuty and Inspector.

Internet → EC2 Web/Application → CloudWatch Agent → CloudWatch Logs/Metrics. Security services operate alongside the workload rather than being mixed into the application path.

3. Monitoring Framework

CPU and EC2 status checks are monitored with CloudWatch alarms. The CloudWatch Agent adds memory and disk metrics and ships Nginx access/error logs and system messages to CloudWatch Logs. This supports both real-time operational monitoring and post-incident troubleshooting.

4. Logging Framework

Application/system logs are centralized in CloudWatch Logs with separate log groups. VPC Flow Logs capture network-flow metadata. CloudTrail records AWS API activity and stores the audit trail in S3 with versioning enabled.

5. Security Architecture

The security group exposes HTTP/HTTPS only for lab validation and intentionally does not expose SSH. EC2 uses an IAM role and Systems Manager. GuardDuty is enabled for threat detection, while Inspector is enabled for EC2 vulnerability assessment. CloudTrail and Flow Logs provide investigation evidence.

6. Alerting

The included CPU alarm triggers above 70% average CPU for two one-minute periods. A status-check alarm detects EC2 status-check failure. Production should add latency, HTTP 5xx, unhealthy-target, storage, database and security-finding alerts.

7. Testing and Validation

```
terraform fmt -check
terraform validate
terraform plan
terraform apply
terraform output
```

After deployment, generate controlled HTTP traffic and confirm Nginx logs arrive in CloudWatch. Run a temporary CPU stress test in the lab and observe the CPU alarm. Confirm CloudTrail events, VPC Flow Logs, GuardDuty status, Inspector status and Security Group rules.

8. Challenges and Resolutions

Default EC2 monitoring does not provide memory/disk metrics, so the CloudWatch Agent is configured. Direct SSH increases exposure, so Systems Manager is used. Operational alarms and security findings are handled by different services because a CPU alarm is not equivalent to a threat-detection finding.

9. Validation Matrix

Test	Expected result	Evidence
Terraform validate	Configuration valid	Terminal
Application	HTTP page returns	Browser
Nginx logs	Events visible in CloudWatch	CloudWatch
CPU stress	Alarm reaches ALARM state	CloudWatch
CloudTrail	API events available	CloudTrail
VPC Flow Logs	Traffic records visible	CloudWatch
GuardDuty	Detector enabled	GuardDuty
Inspector	EC2 assessment enabled	Inspector
Security Group	No SSH/22 ingress	AWS Console

10. Post-deployment Analysis

The design separates operational monitoring from security detection: CloudWatch for workload health, CloudTrail for audit, VPC Flow Logs for network investigation, GuardDuty for threat detection and Inspector for vulnerability assessment. The trade-off is additional cost and configuration complexity, so production retention and alerting policies should be deliberate.

11. Limitations and Production Improvements

This is a lab simulation using a simple Nginx application and HTTP for straightforward validation. Production should use HTTPS/ACM, WAF, private application instances behind an ALB, Multi-AZ architecture, centralized SIEM integration, stronger IAM controls and tested incident-response procedures.

12. Conclusion

The implementation establishes a practical baseline for monitoring, logging and cloud security. Genuine deployment screenshots and test results should be appended before submission.

13. References

AWS documentation: Amazon CloudWatch, CloudWatch Agent, CloudTrail, VPC Flow Logs, Amazon GuardDuty, Amazon Inspector, IAM, Systems Manager and Amazon S3.